

Data Platform (On-Prem)

Problem Statement

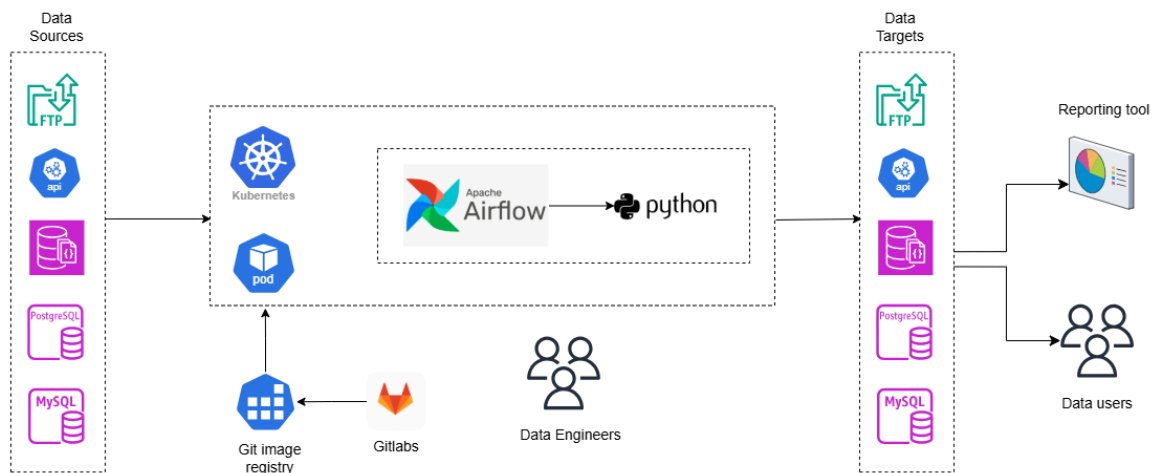
To build a Platform used to design customized and scalable ETL Data Pipeline implementing Medallion architecture to make data readily available for BI reporting

Tech Stack

Apache Airflow | Docker | Kubernetes | Python/SQL

Overview

Data Platform is a scalable solution built to design and orchestrate data pipelines across structured and semi-structured data sources. Using a medallion architecture and Apache Airflow for workflow management, it ensures reliable data movement, transformation, and version-controlled schema management within modern data environments.



Architecture Design of Data Platform

Description

The Data Platform is designed to support scalable, maintainable, and robust data pipelines using a medallion architecture approach. It seamlessly integrates multiple data sources—including both structured databases (such as MySQL, MSSQL, and PostgreSQL) and semi-structured sources (like MongoDB and RESTful APIs)—into various target systems.

The platform employs **batch processing** workflows that are orchestrated using **Apache Airflow**, ensuring efficient scheduling and monitoring of data pipelines. For structured databases, schema migrations are handled through **Alembic**, enabling version-controlled upgrades. Semi-structured databases, like MongoDB, are managed using appropriate tools such as **Mongo Shell** for schema modifications and data interaction.

Development and deployment are containerized using **Docker**, leveraging the official Apache Airflow image. The solution is version-controlled on **GitLab**, where container images are stored in a private registry. For production, the platform is deployed on **Kubernetes**, offering scalability and reliability in enterprise-grade environments

CRM ETL & Warehouse Solution

Problem Statement

To develop an ETL pipeline that extracts data from various HubSpot users and stores it in an AWS Data Lake (S3) utilizing Airbyte, along with constructing a data warehouse for effective data querying and management.

Tech Stack

Python Flask | Airbyte OSS (Open Source Software) | AWS EC2, S3, Glue, Athena

Overview

The CRM ETL & Warehouse solution consists of a HubSpot application developed using the Python Flask framework which is installed by multiple users to extract and process targeted user data. This pipeline leverages Airbyte connectors for data extraction, employs AWS Glue for transformation, and utilizes various AWS services (S3, Glue, Athena) for storage and reporting. Figure 2 illustrates the architecture diagram of the whole pipeline

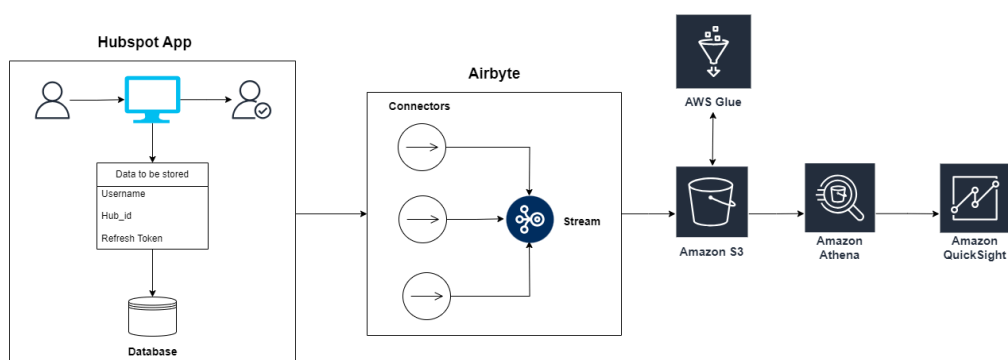


Figure 2: Architecture Design of Brainvoy ELT pipeline

Description

The application, built with Flask in Python, integrates OAuth 2.0 to enable users to authenticate with their HubSpot accounts. Upon authentication, a refresh token is generated for each user, which Airbyte OSS (hosted on AWS EC2) uses to create individual connectors, allowing data extraction from HubSpot and storage in Amazon S3. Each connector is configured with unique user credentials (username, Hub ID, refresh token) to retrieve data across various scopes, such as companies and deals, ensuring customized data synchronization. Data in S3 is stored in user-specific Parquet files, organized by Timeline (YEAR/MONTH/DAY) for accurate storage. AWS Glue jobs clean the data, and crawlers load it into the AWS Glue Catalog, where it can be queried using Athena, supporting the creation of a data warehouse.

Real Estate ETL & BI Solution

Problem Statement

To develop an AWS hosted ETL and BI solution to monitor the live stream data coming from Hospi Housing Application

Tech Stack

AWS Kinesis, Firehose, Lambda, S3, Glue, Athena, QuickSight

Overview

Hospi Housing is an application that connects hosts who want to offer their houses for living to students seeking accommodation in the Netherlands. The application collects various types of data about hosts, students, and their interactions, storing it in DynamoDB, which is configured in AWS. Our task was to set up an ETL pipeline to extract data from DynamoDB, transform it into the correct format, and make it available for visualization. To accomplish this, we utilized several AWS services, including S3, Glue, Athena, and QuickSight. Figure 3 shows the workflow of our ETL pipeline

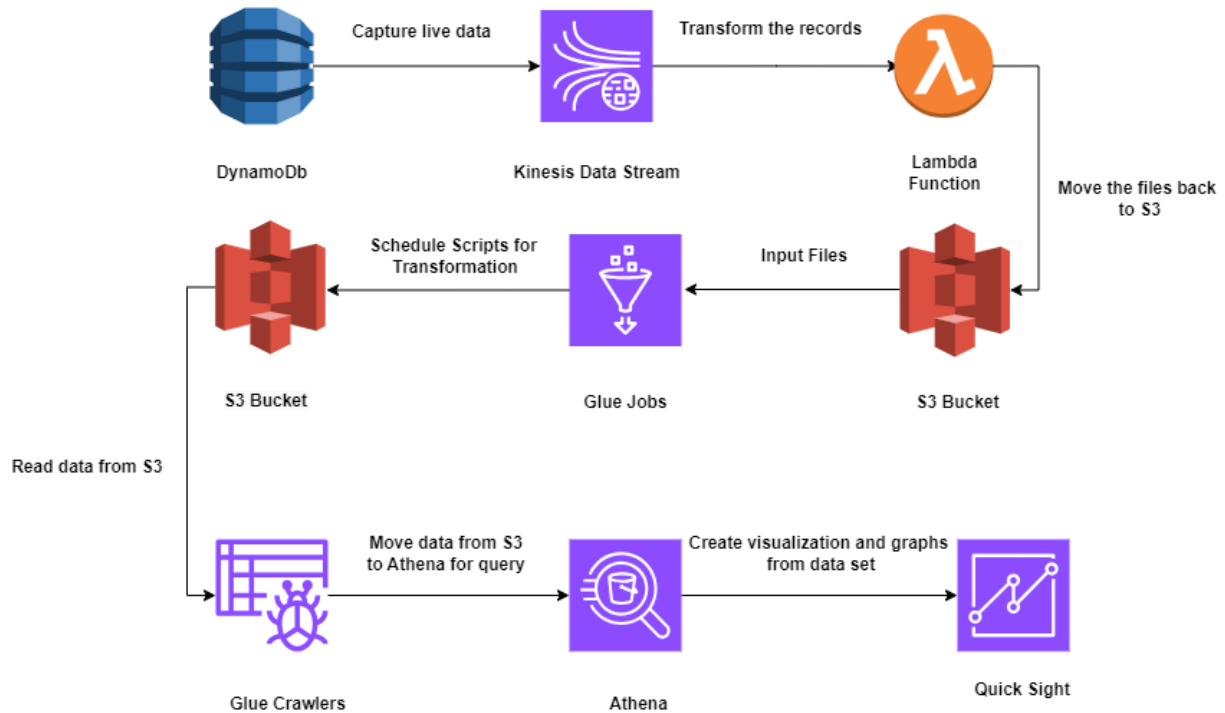


Figure 2: Architecture Design of Hospi Housing ETL/BI

Description

Hospi Housing is an application designed to connect hosts offering housing in the Netherlands with students seeking accommodations. It uses DynamoDB on AWS to store data on hosts, students, and their interactions. To support data analysis, an ETL pipeline was developed to extract data from DynamoDB, transform it into a suitable format, and prepare it for visualization. Key AWS services such as S3, Glue, Athena, and QuickSight were employed in this pipeline. Historical data is exported as JSON files to S3, while live data is streamed hourly via Kinesis and stored in Parquet format. Transformations are handled by Glue, where Python scripts clean and consolidate data daily, and additional SQL queries in Athena help refine the data for visualization. Final data views created in the Glue catalog are integrated with QuickSight, where interactive dashboards provide valuable insights using time-series graphs, maps, and other visualizations.

Meal Project ETL & BI Solution

Problem Statement

To build an effective cloud based solution that will automatically processes the incoming data coming in csv format and provide visual insights

Tech Stack

Azure Blob Container | FunctionApp | SQL Server | PowerBI

Overview

The Meal Project is a cloud-based ETL solution hosted on Azure. It efficiently manages raw data stored in a blob container, processes the data using Function Apps, and ingests the results into a SQL Server database. This processed data is then connected to Power BI, where a report is generated to provide valuable insights. Figure 1 illustrates the architecture diagram of the pipeline.

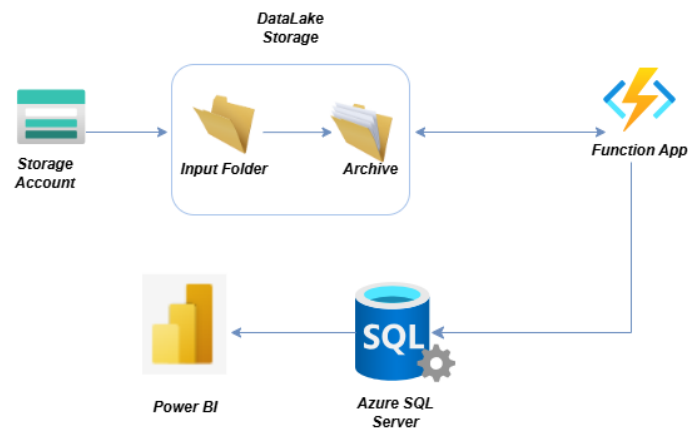


Figure 1: Architecture Design of Meal Project ETL

Description

Raw data submitted by users is stored in an Azure Blob container using Azure Data Storage Explorer, which provides a user-friendly interface for easily uploading files to the desired folder. A Function App, a serverless service, is deployed to trigger at specified intervals. Its role is to retrieve the raw data from the designated folder, process it, and ingest the results into an Azure SQL Database. Once the files are processed, they are archived within the same Blob container. This database is then connected to Power BI, enabling the creation of reports that provide meaningful insights derived from the data.

Amazon ASIN Research Tool

Problem Statement

Develop an automated software tool designed to generate optimized SEO keywords for product ASINs.

Tech Stack

Python Selenium | ChatGPT API

Overview

This project involves gathering and analyzing keywords related to an Amazon product, using ASIN data and a combination of keyword research tools for better market insights. By leveraging SellerApp's free subscription tier, we begin by obtaining a set of primary keywords tied to the product type. These keywords are then expanded through Google Keyword Planner to build a larger, more comprehensive list of related keywords and sent to ChatGPT to get the best keywords among them

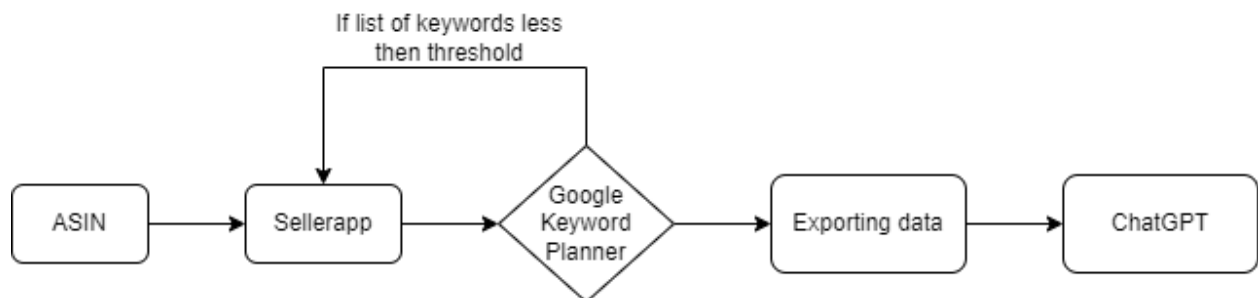


Figure 2: Workflow of Amazon ASIN Research tool

Description

Using Python selenium automation, the ASIN for an Amazon product is first inputted into SellerApp, which generates a foundational set of 10-20 keywords. These keywords provide insight into the product's market positioning and likely search terms. Each keyword is then submitted to Google Keyword Planner, where we obtain around 200-300 related keywords per entry, accumulating until we reach a target list of 2000-3000 keywords. This collection is saved in a .csv file and transferred to ChatGPT using OpenAI API. ChatGPT is then tasked with identifying the top 1000 keywords from this list, as well as the 10 most relevant based on specified criteria. This methodical approach ensures that the final output includes high-value keywords for strategic use.